higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 2

(12010042)

18 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL drawings must be neat, well balanced, reasonably large and fully labelled.
5. ALL the abbreviations and symbols must comply with the latest National Building Regulations and all relevant SANS (SABS) codes.
6. Start each question on a NEW page.
7. Write neatly and legibly.

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QUESTION 1: TERMINOLOGY AND COMPONENTS OF THE CONSTRUCTION INDUSTRY

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 In the contract documents of a construction project, which document tells the contractor how each item should be constructed?

A Contract
B Bill of quantities
C Specifications ☼
D Tender

1.2 Which organisation protects housing consumers against any failures or non-compliance by builders?

A SAICE
B SANS
C NBR
D NHBRC

1.3 Roof tiles are placed on top of … .

A ceiling
B battens ☼
C tie beam
D facia

1.4 The space on a premises, which the municipality keeps for future use where one cannot build, is called ….

A the boundary line
B the building line
C a servitude
D an edge ☼

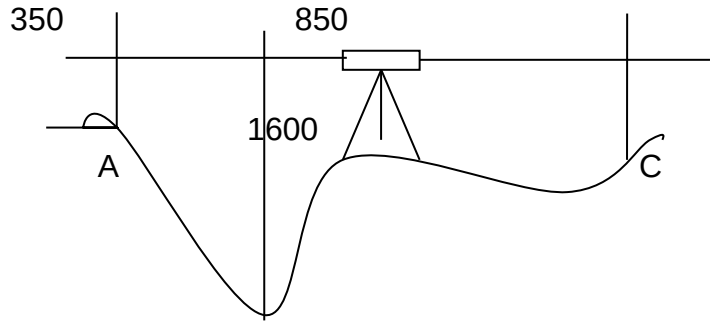
1.5 When building, you are allowed slight variations from the standard, which is called … .

A variation
B tolerance
C extra
D standard

(5 × 1) **[5]**

QUESTION 2: MEASURING, SETTING OUT AND LEVELLING

2.1 FIGURE 1 shows the staff readings from a levelling instrument for points A, B and C. The height of BM B above mean sea level is given as 1 000.



☼

B = BM = 1 000 MSL (Elevation)

FIGURE 1

2.1.1 Calculate the elevation of points A and C from MSL. (4)

2.1.2 Calculate the height difference between A and C. (2)

2.1.3 Briefly describe a *parallax error.* (1)

2.2 State THREE advantages of using a stop or lock button found on a retractable measuring tape. (3)

2.3 A building is set out to clearly define the outline of the excavation and the centre line of the walls, so that construction can be carried out exactly according to the plan.

☼

List the instruments that are used when setting out a residential house. (5)

2.4 What are the limitations of using the 3:4:5 method to set out shapes? (3)

2.5 What is the easiest way to check the squareness of a rectangular set out? (2)

2.6 Explain how you would find the height of a building using a theodolite. (5)

[25]

QUESTION 3: FOUNDATIONS

3.1 Define a *buoyed-up foundation*. (2)

3.2 Describe the working principles of a buoyed-up foundation. ☼ (3)

3.3 Briefly explain under which circumstances buoyed-up foundations are used. (3)

3.4 Sate the width and thickness of a concrete strip footing foundation of a single storey house as prescribed by national standards. (2)

[10]

QUESTION 4: BEHAVIOUR OF STRUCTURES

4.1 Complete the following sentences by filling the missing word or words. Write only the word or words next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

4.1.1 The stress that tries to push the particles of a structure closer together is called … .

4.1.2 The change in shape of a body caused by stress is called … .

4.1.3 The area inside a load carrying beam that experiences neither compression nor tension is called … . ☼

4.1.4 Steel reinforcing used in the upper part of a beam or a slab is to resist … .

4.1.5 The load that tends to cut a component is called … .

(5 × 1) (5)

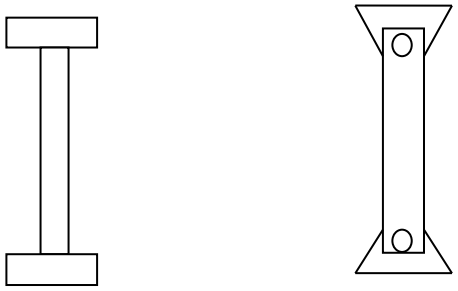
4.2 Name and explain the TWO kinds of failures that occur when different types of loads are applied on top of the arches. (2 × 3) (6)

4.3 Briefly discuss a pin-jointed structure. (2)

4.4 The effective length of a column is defined as the distance between successive inflection points or points of zero moment. ☼

FIGURE 2 below indicates TWO columns. ONE is fixed at both ends and the other is pinned at both ends. Using freehand drawing, show the effective length of each column.

☼



Both ends fixed Both ends pinned

FIGURE 2 (4)

4.5 Define the following construction terminology:

4.5.1 Point load ☼

4.5.2 Uniform distributed load

4.5.3 Non-uniformly distributed load

4.5.4 Axial load

(4 × 2) (8)
[25]

QUESTION 5: INTERNATIONAL SYSTEM (SI) UNITS OF MEASUREMENTS

5.1 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–E) next to the question number (5.1.1–5.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
5.1.1	Electric current	A	candelas
5.1.2	Luminous intensity ☼	B	degrees
5.1.3	Angle	C	Newton
5.1.4	Temperature	D	ampere
5.1.5	Force	E	Kelvin

(5 × 1) (5)

5.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (5.2.1–5.2.5) in the ANSWER BOOK.

5.2.1 100 hectometers = 10 000 meters

5.2.2 50 micrometers = 0,00005 meters

5.2.3 5 kPa = 5 000 N.m² ☼

5.2.4 The unit for density is kg/m²

5.2.5 35 minutes = 125 000 milliseconds

(5 × 1) (5)
[10]

QUESTION 6: AREAS AND VOLUME

6.1 Mr Mole has wall of 5 000 m² that needs to be painted in five days. ☼
How many painters will Mr Mole need to employ if one painter covers 250 m² in a day? (3)

6.2 The plan view of a house is given below.

Calculate the volume in m³ of concrete needed for a concrete strip foundation footing for the entire house. The concrete strip foundation footing has a width of 600 mm and a thickness of 200 mm. ☼

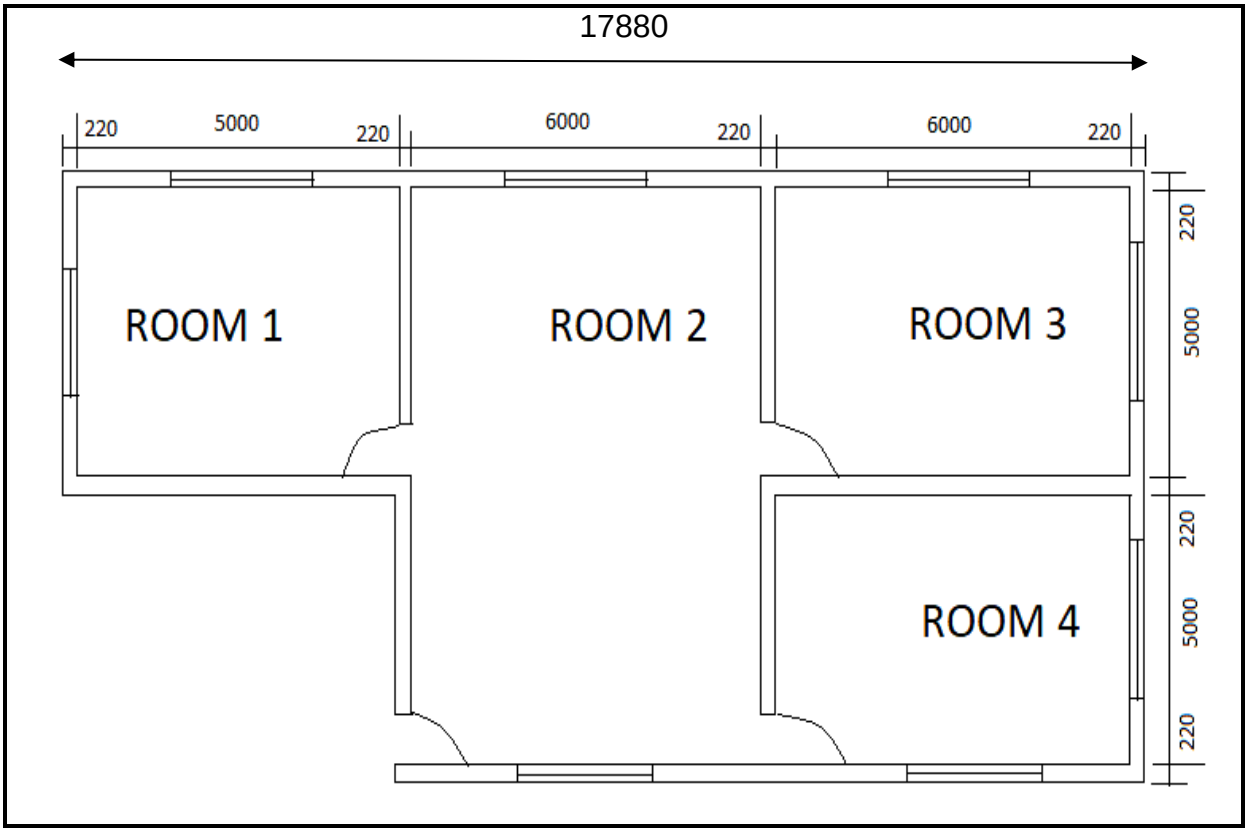


FIGURE 3

(7)
[10]

QUESTION 7: TECHNICAL DRAWINGS

7.1 What does the following abbreviations used in the construction industry stand for?

7.1.1 SWD

7.1.2 PVC ☼

7.1.3 Ms

7.1.4 NGL

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7.1.5 WC

(5 × 1) (5)

7.2 Why is it important to check and inspect new drawings before the building of the structure starts? (2)

7.3 FIGURE 4 shows a vertical section of a cavity wall with a wooden window frame. Identify the different materials represented by the alphabet letters in the drawing. Write only the answer next to the letter (A–E) in the ANSWER BOOK. ☼

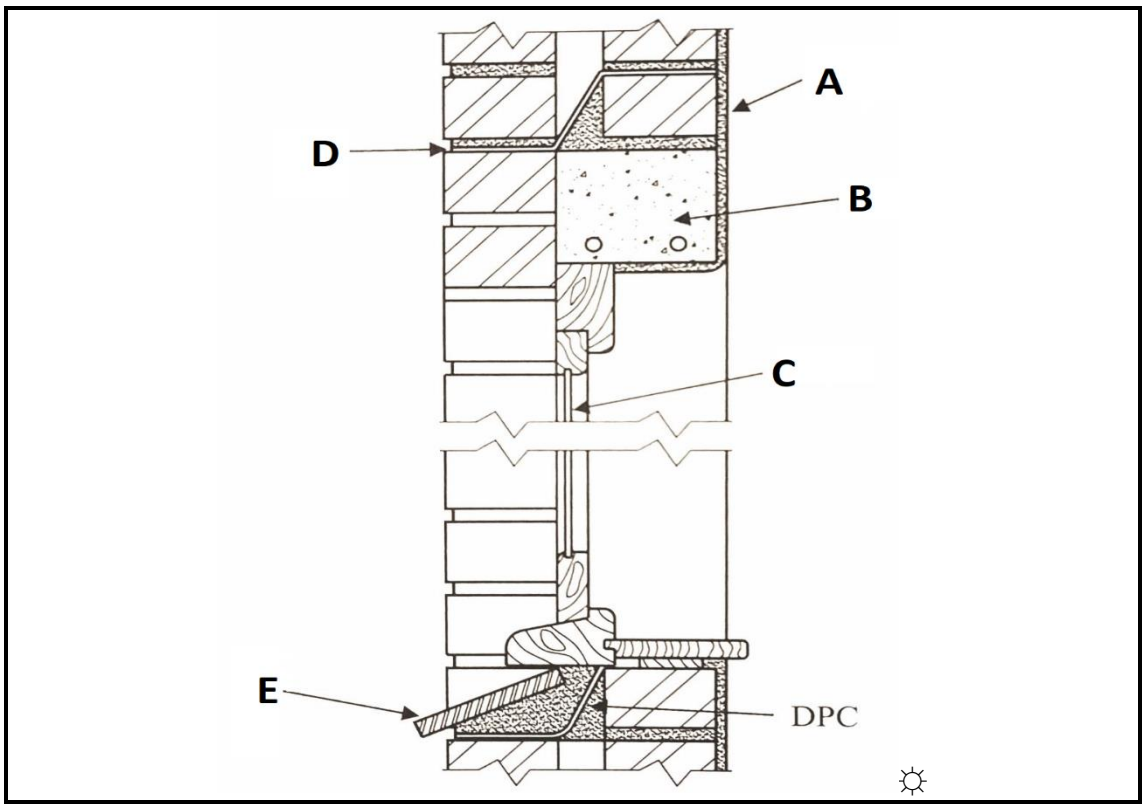


FIGURE 4 (5)

7.4 Explain the following drawing terms:

7.4.1 Plan

7.4.2 Section ☼

7.4.3 Elevation

(3 × 1) (3)
[15]

TOTAL: 100

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